GRADE 7 ACCELERATED · UNIT 4 · LESSON 4

Real-World Problems and Significant Digits

Question Bank 4.04 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, give the number of significant digits in each measurement.

1. 4,207 grams

4

Every digit counts: the zero sits between two nonzero digits.

2. 0.0058 meters

2

Leading zeros are never significant.

3. 3.200 seconds

4

The trailing zeros count, because a decimal point is written. 3.200 and 3.2 are the same number but not the same claim.

4. 52,000 litres

2

No decimal point is written, so the trailing zeros are only holding places.

For questions 5 through 8, solve. Report each answer with the correct precision.

5. A rectangular plot measures 3.5×10^2 m by 1.24×10^2 m. Find its area.

4.3×10^4 ***square meters***

43,400 exactly; 2 significant digits and 3, so the answer keeps 2.

6. A car travels 2.60×10^2 km in 3.0 hours. Find its average speed.

8.7×10^1 ***km/h***

260 ÷ 3.0 = 86.66…; 3 significant digits and 2, so the answer keeps 2.

7. Two lengths are measured: 8.7×10^2 mm and 4.5×10^2 mm. Find their sum.

1.32×10^3 ***mm***

870 and 450 both stop at the tens place, so the sum, 1,320, does too. A sum can carry more significant digits than either measurement — three here, from two twos.

8. Two masses are measured: 8.40×10^{-2} g and 3.715×10^{-3} g. Find their difference.

8.03×10^{-2} ***g***

0.080285 exactly; 0.0840 stops at the ten-thousandths place and 0.003715 stops two places further out, so the answer stops at the ten-thousandths.

Real quantities.

9. The Amazon River discharges about 5.5×10^{7} gallons of water each second.

a. How many gallons does it discharge in one hour, which is 3.6×10^{3} seconds? Report the answer with the correct number of significant digits.

2.0×10^{11} ***gallons***

1.98×10^{11} *exactly. The rate carries two significant digits; 3,600 seconds in an hour is an exact conversion, not a measurement, so it does not limit the count. The answer keeps two.*

b. A second river discharges about 1.10×10^{7} gallons per second. How many times larger is the Amazon's rate? Report the answer with the correct number of significant digits.

5.0 times larger

Exactly 5; two significant digits are justified, so it is written 5.0.

10. A processor performs about 4.66×10^{8} calculations each second.

a. How many calculations does it perform in one minute? Report the answer with the correct number of significant digits.

2.80×10^{10} ***calculations***

$4.66 \times 10^{8} \cdot 60 = 2.796 \times 10^{10}$ *. Sixty seconds in a minute is exact, not a measurement, so it does not limit the count — the answer keeps the measurement's three digits.*

b. Five of these processors run for 20 minutes. How many calculations do they perform altogether? Report the answer with the correct number of significant digits.

2.80×10^{12} ***calculations***

$2.796 \times 10^{10} \cdot 5 \cdot 20 = 2.796 \times 10^{12}$ *. Five and twenty are counts, so again only the measurement's three digits survive.*

Reading someone else's work.

11. Omar multiplies a measured length of 2.5×10^{2} m by a measured width of 1.346×10^{2} m and reports the area as 3.365×10^{4} square meters. Is his arithmetic correct? Is his answer correct? Explain.

His arithmetic is correct — the exact product is 3.365×10^{4} ***. His answer is not, because 2.5 carries only two significant digits, so the area can carry only two:*** 3.4×10^{4} ***square meters.***

A correct value in the wrong form is still the wrong answer here; the rule is part of the question.

12. A rectangle's sides are measured as 7.2×10^1 cm and 4.0×10^1 cm. Which is the area, in scientific notation with the correct number of significant digits?

- ○ A. 2.9×10^2 cm^2
- ○ B. 28.8×10^2 cm^2
- ● **C.** 2.9×10^3 **cm^2**
- ○ D. 2.88×10^3 cm^2

C